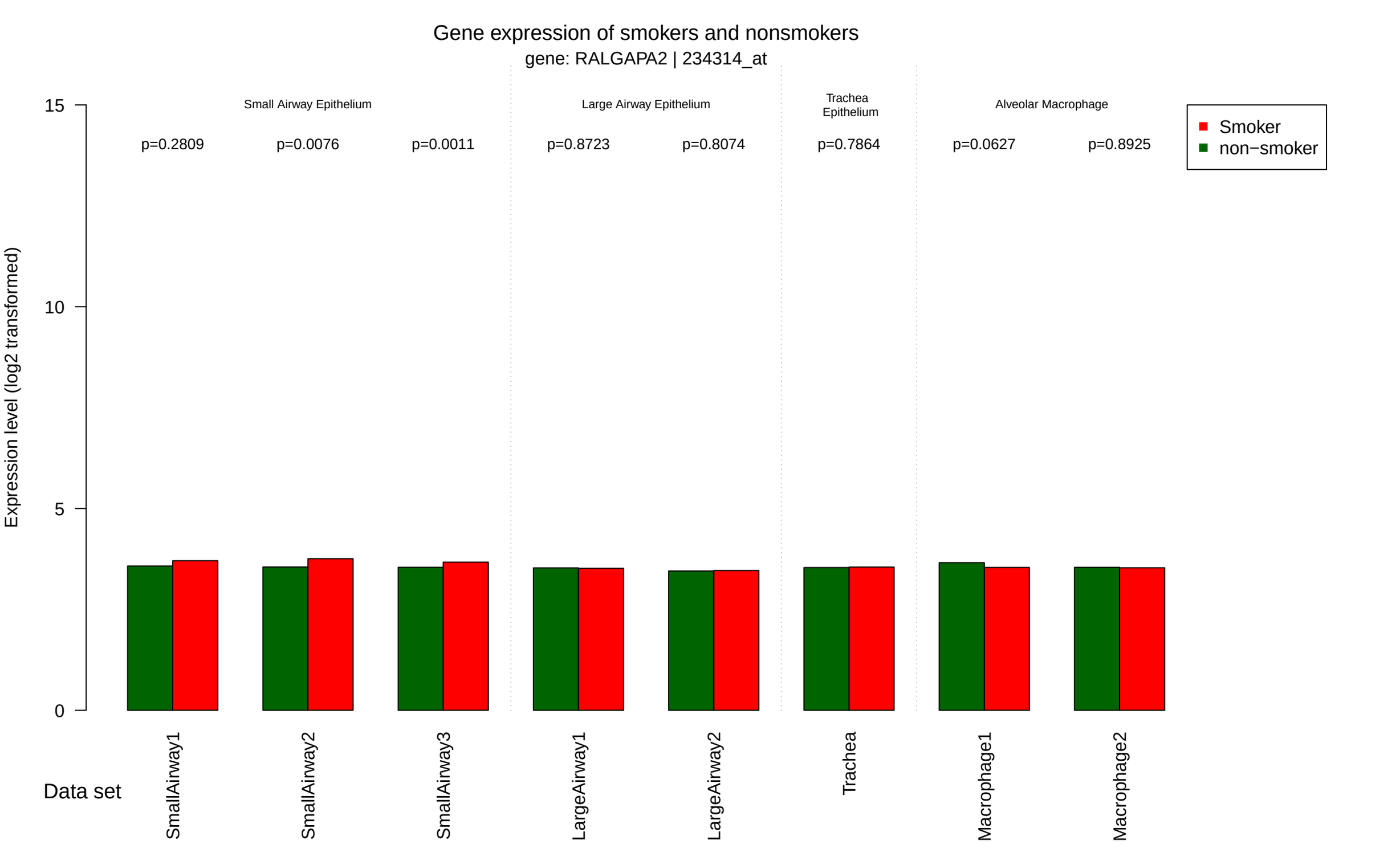
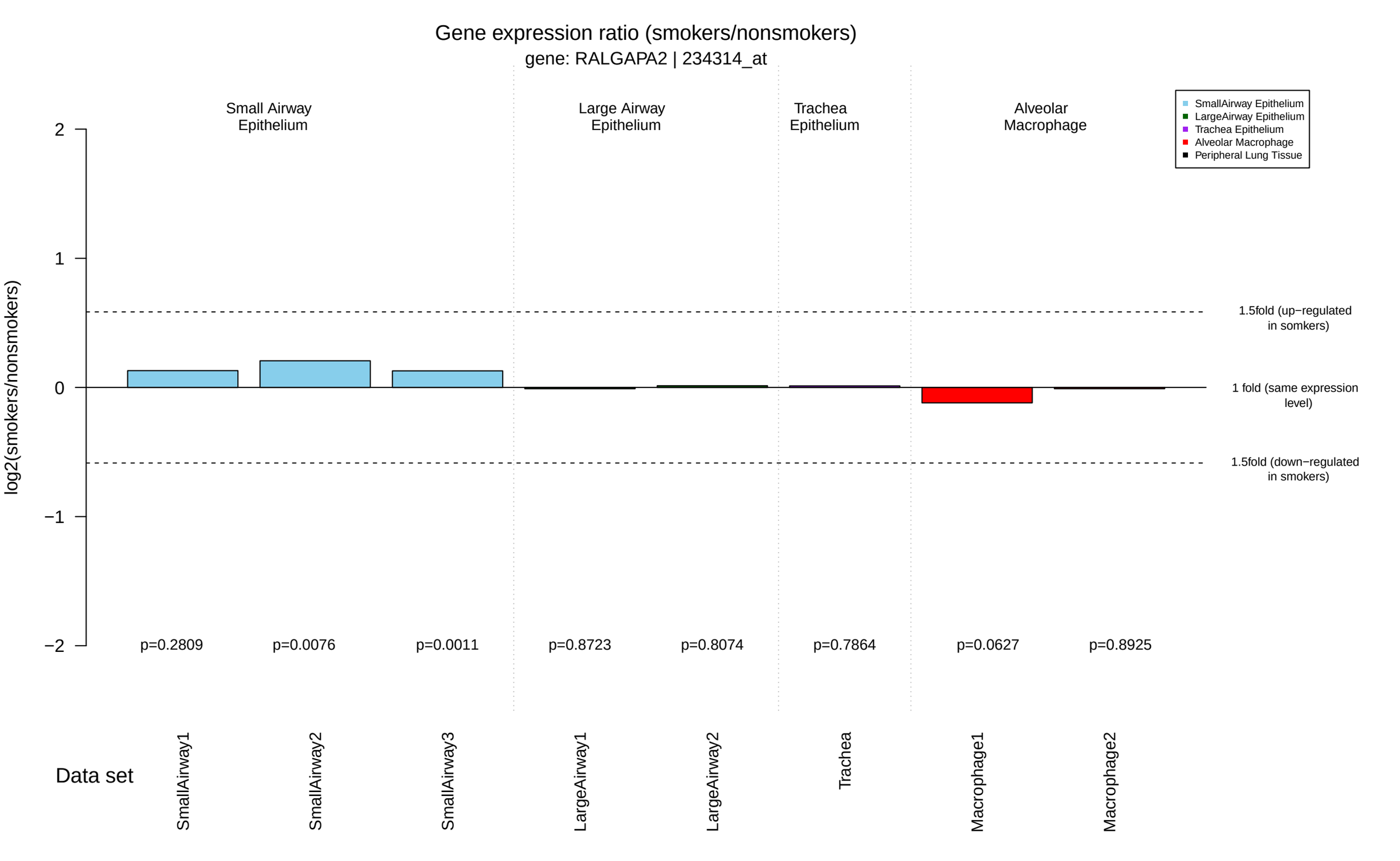
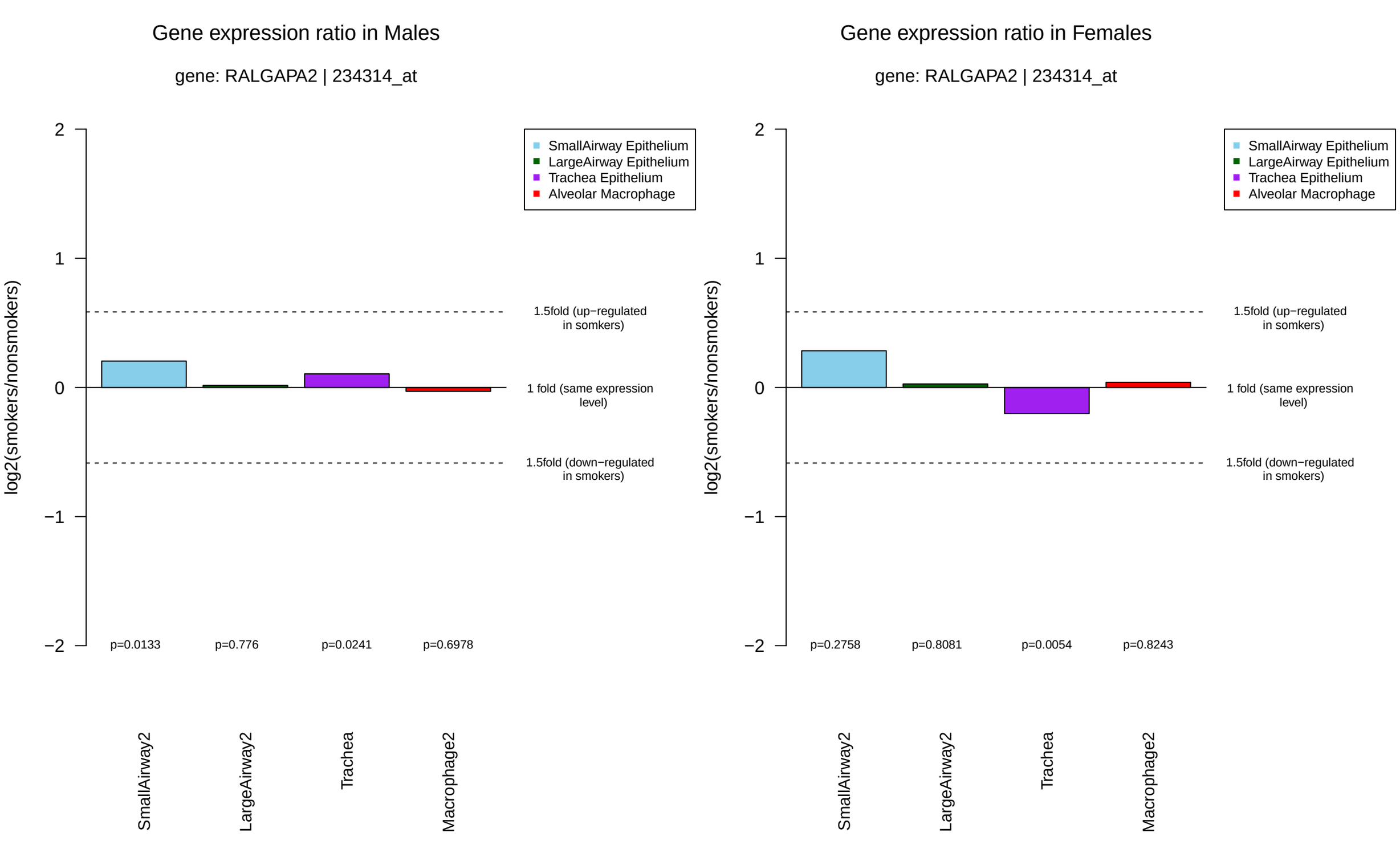
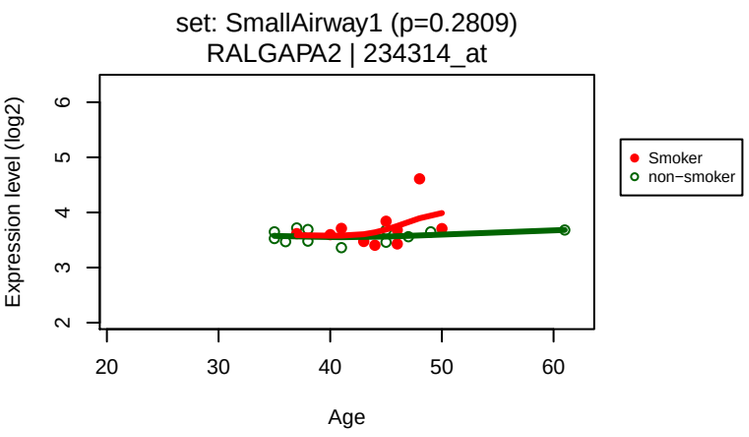
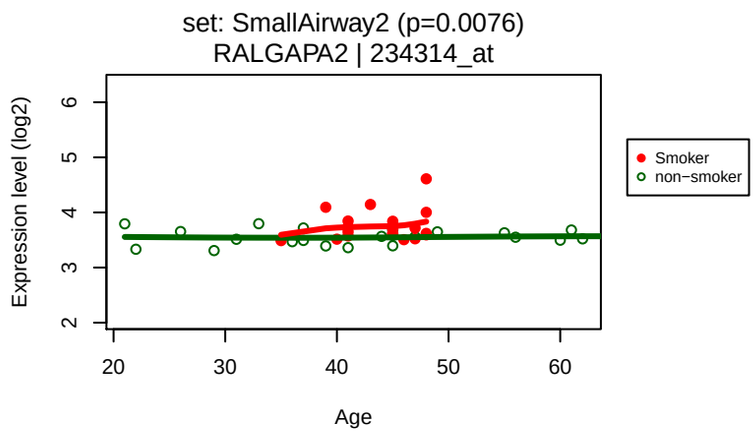
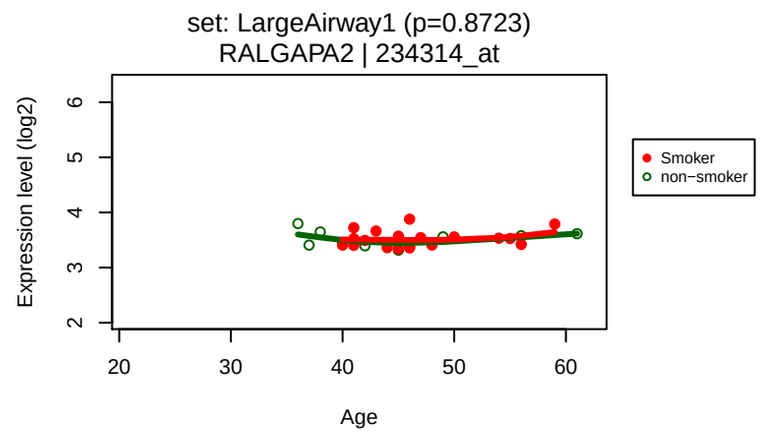

Gene expression of smokers and nonsmokers
gene: RALGAPA2 | 234314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2809
p=0.0076
p=0.0011
p=0.8723
p=0.8074
p=0.7864
p=0.0627
p=0.8925
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

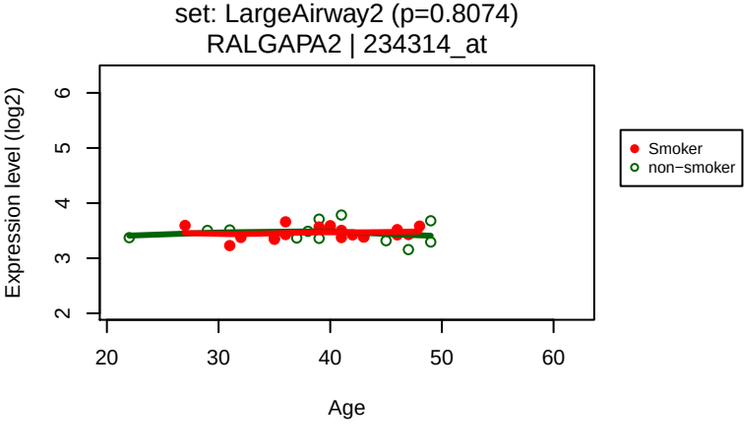

Gene expression ratio (smokers/nonsmokers)
gene: RALGAPA2 | 234314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2809
p=0.0076
p=0.0011
p=0.8723
p=0.8074
p=0.7864
p=0.0627
p=0.8925
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

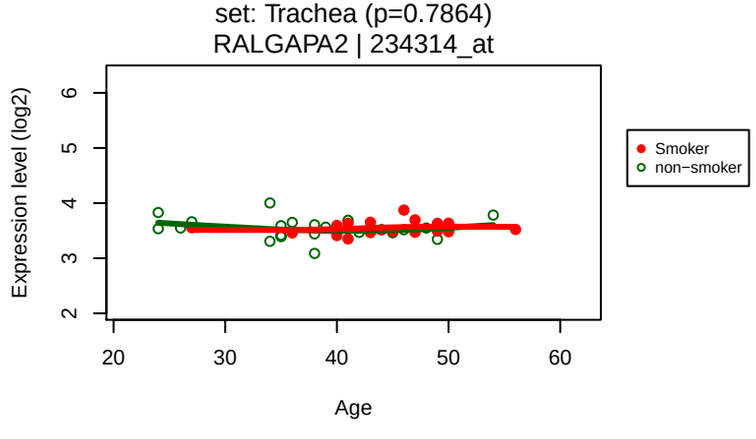

Gene expression ratio in Males
gene: RALGAPA2 | 234314_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0133
p=0.776
p=0.0241
p=0.6978
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGAPA2 | 234314_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2758
p=0.8081
p=0.0054
p=0.8243
SmallAirway2
LargeAirway2
Trachea
Macrophage2

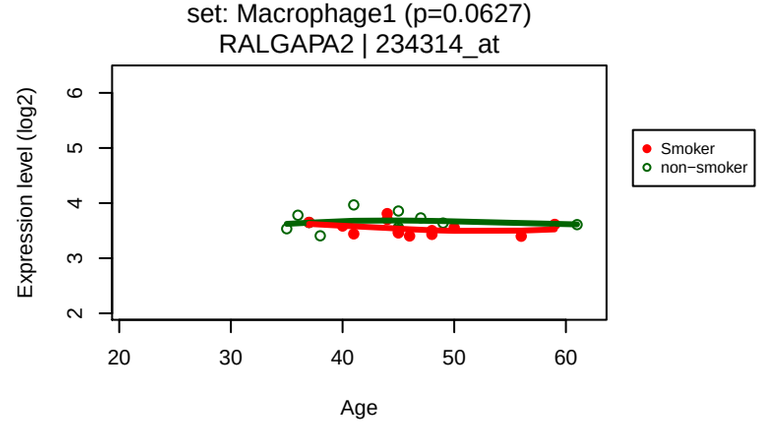

set: SmallAirway1 (p=0.2809)
RALGAPA2 | 234314_at
Expression level (log2)
Age
Smoker
non−smoker

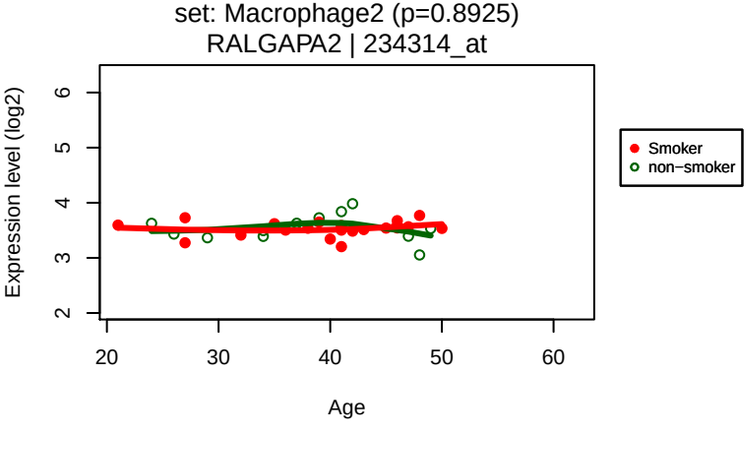

set: SmallAirway2 (p=0.0076)
RALGAPA2 | 234314_at
Expression level (log2)
Age
Smoker
non−smoker

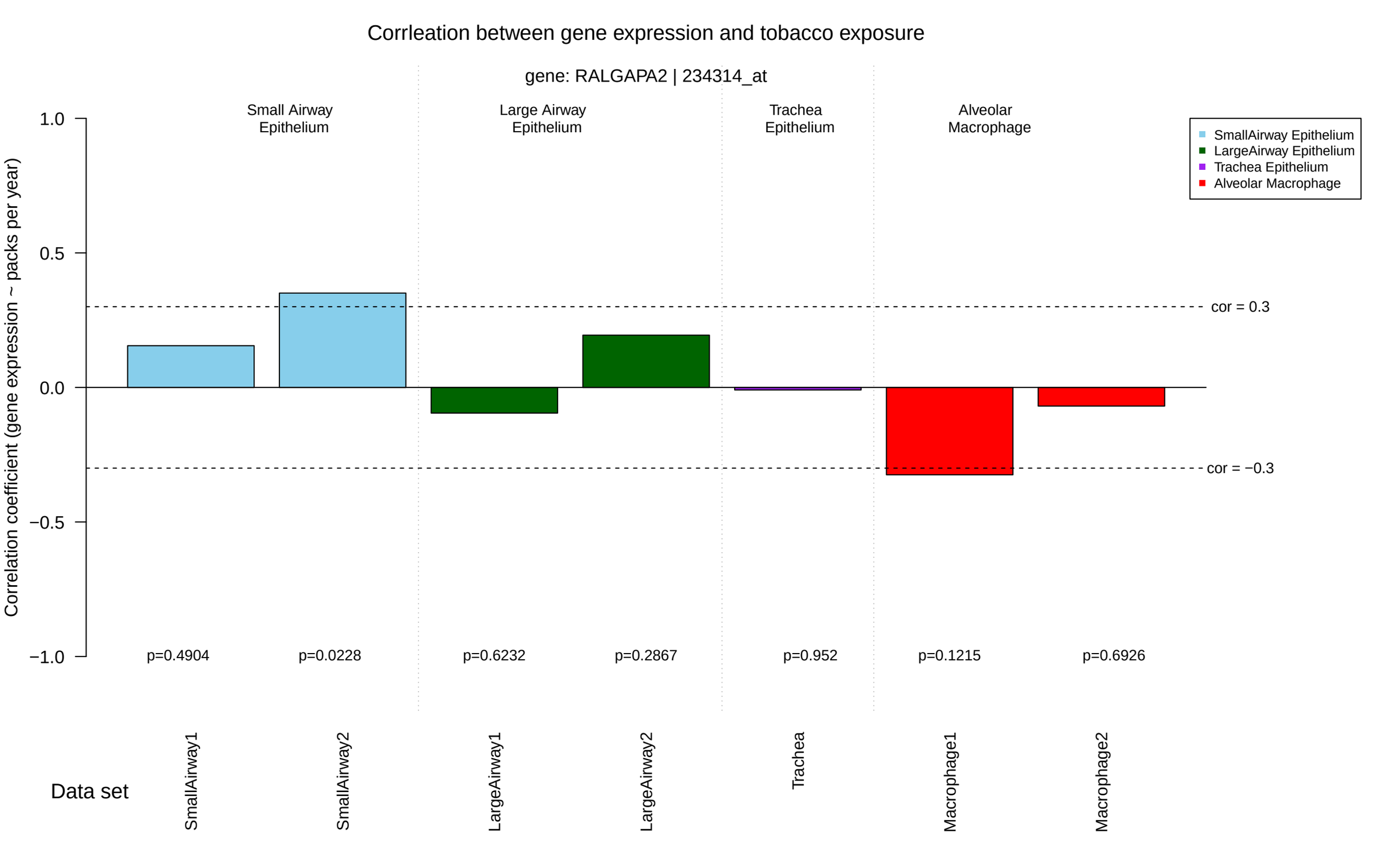

set: LargeAirway1 (p=0.8723)
RALGAPA2 | 234314_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8074)
RALGAPA2 | 234314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7864)
RALGAPA2 | 234314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0627)
RALGAPA2 | 234314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8925)
RALGAPA2 | 234314_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGAPA2 | 234314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4904
p=0.0228
p=0.6232
p=0.2867
p=0.952
p=0.1215
p=0.6926
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2